

Giovanni Luca Marchetti

Postdoctoral Researcher in Theoretical Machine Learning | Email: glma@kth.se | [Webpage](#)
Department of Mathematics, Royal Institute of Technology (KTH), Stockholm, Sweden.

EDUCATION

- Ph.D. Computer Science 05/2020 – 04/2024
Royal Institute of Technology (KTH), Stockholm, Sweden
 - Worked on geometric deep learning and high-dimensional statistics, with applications to robotics.
 - Thesis title: On Symmetries and Metrics in Geometric Inference.
 - Supervisor: Danica Kragic. Co-supervisor: Anastasia Varava.
- Ph.D. (uncompleted) Pure Mathematics 10/2017 – 03/2020
University of Sheffield, United Kingdom
 - Worked on algebraic geometry and category theory, with applications to string theory.
 - Supervisor: Tom Bridgeland.
- M.Sc. Pure Mathematics 10/2015 – 07/2017
University of Rome La Sapienza, Italy
 - Grade: Summa cum laude, with an excellence award.
- B.Sc. Pure Mathematics 10/2012 – 07/2015
University of Rome La Sapienza, Italy
 - Grade: Summa cum laude, with an excellence award.

WORK EXPERIENCE

- Postdoctoral Researcher 05/2024 – Now
Royal Institute of Technology (KTH), Stockholm, Sweden
 - Working on theoretical approaches to deep learning via algebraic geometry.
 - Supervisor: Kathlén Kohn.
- Research Intern 06/2023 – 10/2023
Qualcomm AI Research, Amsterdam, The Netherlands
 - Worked on geometric deep learning for combinatorial optimization, with applications to signal processing.

PUBLICATIONS

- Giovanni Luca Marchetti[†], Erin Connelly[†], Paul Breiding[†], and Kathlén Kohn[†]. *Critical Points of Degenerate Metrics on Algebraic Varieties: A Tale of Overparametrization*. In: Preprint. 2026.
- Vahid Shahverdi[†], Giovanni Luca Marchetti[†], Georg Bökman[†], and Kathlén Kohn[†]. *Identifiable Equivariant Networks are Layerwise Equivariant*. In: International Conference on Machine Learning (ICML). 2026
- Giovanni Luca Marchetti[†], Daniel Kunin[†], Adele Myers, Francisco Acosta, and Nina Miolane. *Sequential Group Composition: A Window into the Mechanics of Deep Learning*. In: International Conference on Machine Learning (ICML). 2026.
- Vahid Shahverdi[†], Giovanni Luca Marchetti[†], and Kathlén Kohn[†]. *Learning on a Razor's Edge: Identifiability and Singularity of Polynomial Neural Networks*. In: International Conference on Learning Representations (ICLR). 2026.
- Aniss Aiman Medbouhi[†], Alejandro García-Castellanos[†], Giovanni Luca Marchetti, Daniel Pelt, Erik J Bekkers, and Danica Kragic. *Randomized HyperSteiner: A Stochastic Delaunay Triangulation Heuristic for the Hyperbolic Steiner Minimal Tree*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2026.
- Daniel Kunin, Giovanni Luca Marchetti, Feng Chen, Dhruva Karkada, James B Simon, Michael R DeWeese, Surya Ganguli, and Nina Miolane. *Alternating Gradient Flows: A Theory of Feature Learning in Two-layer Neural Networks*. In: Advances in Neural Information Processing Systems (NeurIPS). 2025.
- Zheyu Zhuang, Ruiyu Wang, Giovanni Luca Marchetti, Florian T Pokorny, and Danica Kragic. *MirrorDuo: Reflection-Consistent Visuomotor Learning from Mirrored Demonstration Pairs*. In: Annual Conference on Robot Learning (CoRL). 2025.
- Giovanni Luca Marchetti[†], Vahid Shahverdi[†], Stefano Mereta[†], Matthew Trager[†], and Kathlén Kohn[†]. *Algebra Unveils Deep Learning: An Invitation to Neuroalgebraic Geometry*. In: International Conference on Machine Learning (ICML). 2025. **Spotlight Paper**.
- Nathan W Henry[†], Giovanni Luca Marchetti[†], and Kathlén Kohn[†]. *Geometry of Lightning Self-Attention: Identifiability and Dimension*. In: International Conference on Learning Representations (ICLR). 2025.

- Vahid Shahverdi[†], Giovanni Luca Marchetti[†], and Kathlén Kohn[†]. *On the Geometry and Optimization of Polynomial Convolutional Networks*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2025.
- Giovanni Luca Marchetti, Gabriele Cesa, Kumar Pratik, and Arash Behboodi. *Neural Lattice Reduction: A Self-Supervised Geometric Deep Learning Approach*. In: Transactions on Machine Learning Research (TMLR), 2025.
- Alejandro García-Castellanos, Aniss Aiman Medbouhi, Giovanni Luca Marchetti, Erik J Bekkers, and Danica Kragic. *Hypersteiner: Computing Heuristic Hyperbolic Steiner Minimal Trees*. In: Symposium on Algorithm Engineering and Experiments (ALENEX), 2025.
- Giovanni Luca Marchetti, Christopher J Hillar, Danica Kragic, and Sophia Sanborn. *Harmonics of Learning: Universal Fourier Features Emerge in Invariant Networks*. In: Conference on Learning Theory (COLT). 2024.
- Aniss Aiman Medbouhi, Giovanni Luca Marchetti, Vladislav Polianskii, Alexander Kravberg, Petra Poklukar, Anastasia Varava, and Danica Kragic. *Hyperbolic Delaunay Geometric Alignment*. In: European Conference on Machine Learning (ECML-PKDD). 2024.
- Giovanni Luca Marchetti, Vladislav Polianskii, Anastasiia Varava, Florian T Pokorny, and Danica Kragic. *An Efficient and Continuous Voronoi Density Estimator*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2023. **Notable Paper award**.
- Giovanni Luca Marchetti[†], Gustaf Tegnér[†], Anastasiia Varava, and Danica Kragic. *Equivariant Representation Learning via Class-Pose Decomposition*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2023.
- Alfredo Reichlin[†], Giovanni Luca Marchetti[†], Hang Yin, Anastasiia Varava, and Danica Kragic. *Learning Geometric Representations of Objects via Interaction*. In: European Conference on Machine Learning (ECML-PKDD). 2023.
- Luis Perez Rey[†], Giovanni Luca Marchetti[†], Danica Kragic, Dimitri Jarnikov, and Mike Holenderski. *Equivariant Representation Learning in the Presence of Stabilizers*. In: European Conference on Machine Learning (ECML-PKDD). 2023.
- Alexander Kravberg[†], Giovanni Luca Marchetti[†], Vladislav Polianskii[†], Anastasiia Varava, Florian T Pokorny, and Danica Kragic. *Active Nearest Neighbor Regression Through Delaunay Refinement*. In: International Conference on Machine Learning (ICML). 2022.
- Vladislav Polianskii[†], Giovanni Luca Marchetti[†], Alexander Kravberg, Anastasiia Varava, Florian T Pokorny, and Danica Kragic. *Voronoi Density Estimator for High-Dimensional Data: Computation, Compactification and Convergence*. In: Uncertainty in Artificial Intelligence (UAI). 2022.
- Alfredo Reichlin, Giovanni Luca Marchetti, Hang Yin, Ali Ghadirzadeh, and Danica Kragic. *Back to the Manifold: Recovering from Out-of-Distribution States*. In: International Conference on Intelligent Robots and Systems (IROS). 2022.
- Domenico Fiorenza[†], Fosco Loregian[†], and Giovanni Luca Marchetti[†]. *Hearts and Towers in Stable Infinity-Categories*. In: Journal of Homotopy and Related Structures. 2019.

Note: The symbol [†] denotes shared first-authorship.

A complete list of publications is available on Google Scholar [\[LINK\]](#).

PATENTS

- Giovanni Luca Marchetti, Cesa Gabriele, Kumar Pratik, and Arash Behboodi. *Geometric Deep Learning for Lattice Reduction*. US Patent US12413270B2. 2024.

TEACHING EXPERIENCE

Contributed to the following courses:

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|--|---------------------------------|
| · Geometry and Topology for Data Analysis (guest lectures) | Univ. of Trento, 2025 |
| · Groups and Rings (guest lectures) | KTH, 2025 |
| · Advanced Linear Algebra (tutorials, examination) | KTH, 2025 |
| · Artificial Intelligence (tutorials, examination, organization) | KTH, 2022 – 2023 |
| · Computer Vision (tutorials, examination) | KTH, 2023 |
| · Database Technology (tutorials, examination) | KTH, 2021 – 2022 |
| · Scientific Programming in Python (tutorials) | Univ. of Sheffield, 2019 |
| · Mechanics and Fluids (tutorials, examination) | Univ. of Sheffield, 2019 |
| · Analysis and Algebra (tutorials) | Univ. of Sheffield, 2018 – 2019 |

SUPERVISORY AND ORGANIZATIONAL ACTIVITY

Mentored the following students:

· Juan Sepulveda (Ms.Sc., supervision)	2026
· Johan Slettengren (Ms.Sc., supervision)	2025
· Aniss Medbouhi (Ph.D., co-supervision)	2023 – Now
· Nathan Henry (research visit, co-supervision)	2024
· Alejandro García Castellanos (Ms.Sc. and research engineer, co-supervision)	2023 – 2024
· Markus Hector (Ms.Sc., supervision)	2023

Co-organized the following events and activities:

· Workshop: <i>Geometric Deep Learning in Umeå (GeUmetric)</i>	2025
· Reading Seminar: <i>Computer Vision and Deep Learning</i>	2022 – 2023
· Reading Seminar: <i>Geometry and Machine Learning</i>	2021 – 2023

SELECTED INVITED TALKS

· <i>The Algebraic Geometry of Deep Learning</i> EPFL AI Center Center Seminar	Lausanne (Switzerland), 2025
· <i>An Invitation to Neuroalgebraic Geometry</i> SIAM Conference on Applied Algebraic Geometry (AG)	Madison (USA), 2025
· <i>On the Geometry and Optimization of Polynomial Convolutional Network</i> Joint Meeting of Mathematicians (JMM)	Seattle (USA), 2025
· <i>Harmonics of Learning: Universal Fourier Features Emerge in Invariant Networks</i> International Conference on Learning Theory (COLT)	Edmonton (Canada), 2024
· <i>Equivariant Representation Learning in the Presence of Stabilizers</i> European Conference on Machine Learning (ECML-PKDD)	Turin (Italy), 2023
· <i>An Efficient and Continuous Voronoi Density Estimator</i> International Conference on Artificial Intelligence and Statistics (AISTATS)	Valencia (Spain), 2023
· <i>Active Nearest Neighbor Regression Through Delaunay Refinement</i> International Conference on Machine Learning (ICML)	Baltimore (USA), 2022
· <i>Higher Categories in Mirror Symmetry</i> Junior Algebraic Geometry Seminar	Univ. of Warwick (UK), 2019

SELECTED SOFTWARE

Developed the following machine learning software:

- [\[LINK\]](#) A deep self-supervised framework for learning group-equivariant representations (in PyTorch).
- [\[LINK\]](#) A contrastive complex-valued network based on harmonic analysis (in PyTorch, JAX).
- [\[LINK\]](#) A high-dimensional non-parametric density estimator based on Voronoi cells (in C++, NumPy).

Further software is available on GitHub [\[LINK\]](#).

SKILLS

- *Languages:* Italian (native), Russian (native), English (professional), Swedish (intermediate), French (basic).
- *Programming:* Python (NumPy, PyTorch, JAX), C, C++, $\LaTeX$, Bash, SQL.